SSC MTS 2024 question paper with answers

30 September 2024 · Shift 1

81 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 9 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

Atul sold two sacks of rice for ₹19,200 each. On one, he gained 20% and on the other, he lost 20%. Find his gain or loss percentage in the whole transaction.

(A) 7% profit
(B) 8% profit
(C) 5% loss
(D) 4% loss

Q.2

Riya offers 12% discount on all her goods and offers an additional discount of 6% to those customers who pay in cash. How much will a customer have to pay for a fan marked for ₹2,400 if he pays in cash?

(A) ₹2,300.50
(B) ₹2,122.50
(C) ₹1,985.28
(D) ₹1,900

Q.3

Rice worth ₹43/kg and ₹67/kg are mixed with a third variety in the ratio 2 : 1 : 5. If the mixture is worth ₹96/kg, the price (in ₹) of the third variety of rice per kg will be:

(A) 159
(B) 123
(C) 142
(D) 131

Q.6

In a 100-marks test, the average of marks scored by 50 students was found to be 65. Later on, it was discovered that a score of 88 was misread as 68. Find the correct average.

(A) 65.60
(B) 65.40
(C) 65.70
(D) 65.50

Q.7

A thief saw a policeman, and he started running at a speed of 18 m/s. After 12 seconds, the policeman starts running behind the thief at a certain speed. If the policeman covered 864 meters distance to reach the thief, then the speed (in m/s) of the policeman is:

(A) 15
(B) 30
(C) 24
(D) 28

Q.8

Three numbers are in the ratio of 6:7:3 and their LCM is 3654.Their HCF is:

(A) 27
(B) 87
(C) 29
(D) 28

Q.9

Two trains, 148 m and 152 m long, are running towards each other on parallel lines at 62 km/h and 46km/h, respectively. In what time will they be clear of each other from the moment they meet?

(A) 10 sec
(B) 5 sec
(C) 15 sec
(D) 20 sec

Q.11

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 7% per year rate of interest for 6 years.

(A) 1410
(B) 1470
(C) 1510
(D) 1430

Q.13

In a college, 20% of the number of boys are equal to the three-fourth of the number of girls. What is the ratio of boys and girls in that college ?

(A) 5 : 3
(B) 13 : 4
(C) 7 : 6
(D) 15 : 4

Q.14

The ratio of the present age of A and B is 3 : 1. Five years ago, the ratio of their ages was 5 : 1. The age of A (in years) after eight years will be:

(A) 22
(B) 38
(C) 30
(D) 40

Q.15

The average monthly expenditure of a family was ₹8,600 during the first three months, ₹9,750 during the next four months and ₹9,920 during the last five months of a year. If the total savings during the year was ₹65,300, the average monthly income (in ₹) of the family was:

(A) 14,597
(B) 15,794
(C) 15,479
(D) 14,975

Q.16

A sum invested at 5% simple interest per annum grows to ₹806 in 6 years. If the amount remains the same, how much will it grow at a 12% simple interest rate in 5 years?

(A) ₹1,120
(B) ₹992
(C) ₹1,000
(D) ₹985

Q.17

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 7% per year rate of interest for 5 years.

(A) 1090
(B) 1010
(C) 990
(D) 1050

Q.18

The average monthly income of A and B is ₹3,590. The average monthly income of B and C is ₹4,250. The average monthly income of A and C is ₹6,500. What is the monthly income of C?

(A) ₹7,545
(B) ₹6,984
(C) ₹8,190
(D) ₹7,160

Q.19

A man sells an article at a certain price and suffers a loss of 20%. If he sells the article at ₹100 more, he gets a profit of 30%. Find the cost price of the article (in ₹).

(A) 200
(B) 400
(C) 300
(D) 250

Reasoning Ability and Problem Solving 16 questions

Q.1

What should come in place of ? in the given series based on the English alphabetical order? ACE ,FHJ, KMO, PRT , ?

(A) BCD
(B) NOP
(C) GJK
(D) UWY

Q.2

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the center (but not necessarily in the same order). C is sitting to the immediate right of E. D is sitting third to the left of C. A is the immediate neighbour of B and C. F is not the immediate neighbour of E. What is the position of G with respect to A?

(A) To the immediate left
(B) Second to the left
(C) Second to the right
(D) Third to the left

Q.3

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). A is sitting second to the right of E. Only one person is sitting between D and A. B is sitting third to the right of G. F is an immediate neighbour of B. How many people are sitting between D and E when counted from the left of E?

(A) One
(B) Two
(C) Four
(D) Three

Q.4

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some chickens are hens. No hen is an egg. Conclusions: (I) No chicken is an egg. (II) All hens are chickens.

(A) Both conclusions I and II follow
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Only conclusion I follows

Q.5

HFLI is related to OMSP in a certain way based on the English alphabetical order. In the same way, KIOL is related to RPVS. To which of the following is MKQN related, following the same logic?

(A) RTUX
(B) RTXU
(C) TRXU
(D) TRUX

Q.6

What should come in place of '?' in the given series? -12, +24, -48, +96, ?, +384

(A) -192
(B) 96
(C) -108
(D) 108

Q.8

In a code language, 'COLONY' is coded as '48' and 'INTERCHANGE' is coded as '88'. How will 'MOUNTAIN' be coded in the same language?

(A) 66
(B) 84
(C) 64
(D) 86

Q.9

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: No bus is a car. No car is a truck. Conclusions: (I) No car is a bus. (II) No truck is a car.

(A) Both conclusions I and II follow.
(B) Only conclusion I follows.
(C) Neither conclusion I nor II follows.
(D) Only conclusion II follows.

Q.10

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 39 ÷ 3 – 24 + 60 × 12 = ?

(A) 163
(B) 190
(C) 86
(D) 136

Q.11

In a certain code language, 'WING' is coded as '2618' and 'NEAT' is coded as '3469'. What is the code for 'N' in that language?

(A) 1
(B) 3
(C) 6
(D) 8

Q.12

JM12 is related to EH15 in a certain way. In the same way, QT34 is related to LO37. To which of the following is SV 23 related following the same logic?

(A) MP26
(B) NP27
(C) NQ26
(D) MQ27

Q.14

7 is related to 30 following a certain logic. Following the same logic, 14 is related to 100. To which of the following is 22 related following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 170
(B) 180
(C) 190
(D) 189

Q.15

In a certain code language, 'I swim fast' is coded as 'cp lk bs' and 'may I go' is coded as 'ap sb lk'. How is 'I' coded in that code language?

(A) cp
(B) lk
(C) sb
(D) bs

Q.16

In a code language, 'CAKE' is written as 'ACIG' and 'GOLD' is written as 'EQJF'. How will 'WIRE' be written in that language?

(A) UKPG
(B) GUKP
(C) GPUK
(D) GUPK

Q.17

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13, can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (4, 35, 11) (1, 82, 61)

(A) (2, 53, 6)
(B) (13, 53, 20)
(C) (8, 33, 15)
(D) (21, 310, 23)

Q.19

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 5 ÷ 6 × 3 – 4 + 11 = ?

(A) 7
(B) 14
(C) 11
(D) 3

General Awareness 25 questions

Q.1

In which state/union territory is Losar celebrated to mark the start of the new year in that region?

(A) Kerala
(B) Ladakh
(C) New Delhi
(D) Bihar

Q.2

Who was the captain of Royal Challengers Bangalore in the Women's Premier League event 2023?

(A) Harmanpreet Kaur
(B) Shafali Verma
(C) Deepti Sharma
(D) Smriti Mandhana

Q.3

As of 31st October 2023, the senior-most member (in terms of age) in the 17th Lok Sabha, Shafiqur Rahman Barq represents which political party?

(A) Samajwadi Party
(B) Bharatiya Janata Party
(C) Indian National Congress
(D) Rashtriya Janata Dal

Q.4

With reference to medieval Indian architecture style, what is 'Pietra Dura'?

(A) Decorating the walls with floral designs made of semi-precious stones
(B) Oil paintings on the walls of Taj Mahal
(C) Fine quality clothes used by the Mughals
(D) Sculpture art of the Mauryan period

Q.5

What is the chemical formula for sodium bicarbonate?

(A) NaOH
(B) NaCl
(C) $NaHCO_3$
(D) Na_2CO_3

Q.6

How many countries share their land borders with India?

(A) Nine
(B) Seven
(C) Eight
(D) Six

Q.7
Which of the following books is written by Dadabhai Naoroji?

(A) Economics History of India
(B) Poverty and Un-British Rule in India
(C) Poverty and India
(D) Important Chapters of Poverty

Q.8
The chemical compound present in turmeric is:

(A) curcumin
(B) capsaicin
(C) carotene
(D) tartaric acid

Q.9
Which of the following dynasties established the Vijayanagara kingdom in fourteenth century?

(A) Sangama dynasty
(B) Saluva dynasty
(C) Tuluva dynasty
(D) Aravidu dynasty

Q.10
With which sport is Sanjana Bathula associated?

(A) Roller skating
(B) Shooting
(C) Squash
(D) Badminton

Q.11
The historical fiction 'The Glory of Patan' by KM Munshi was originally written in which Indian language?

(A) Bengali
(B) Tamil
(C) Malayalam
(D) Gujarati

Q.12
The Rabi season of agriculture begins in October-November and ends in _____.

(A) May-June
(B) June-July
(C) March-April
(D) January-February

Q.13
On which date did Punjab abolish the Legislative Council?

(A) 1 January 1971
(B) 1 January 1975
(C) 1 January 1970
(D) 1 January 1977

Q.14
Which Union Territory had minimum literacy rate as per 2011 Census?

(A) Chandigarh
(B) Lakshadweep
(C) Delhi
(D) Dadra and Nagar Haveli

Q.15
Kanak Rele, who established Nalanda Dance Research Centre, was a performer of which of the following classical dance forms of India?

(A) Mohiniyattam
(B) Manipuri
(C) Kathak
(D) Sattriya

Q.16
Who among the following established the Indore gharana?

(A) Ustad Ghaghe Khuda Baksh
(B) Tansen
(C) Ustad Amir Khan
(D) Bhimsen Joshi

Q.17
The red colour of tomatoes is due to the presence of:

(A) limonene
(B) beta carotene
(C) alizarin
(D) lycopene

Q.18
Which of the following is one of the features of capital goods?

(A) They are durable.
(B) They are Giffen goods.
(C) They are inferior goods.
(D) They are inexpensive.

Q.19
In 1976, in which Part of the Indian Constitution under Article 51A were Fundamental Duties added?

(A) IV A
(B) III
(C) VI
(D) V

Q.20
What is the decreasing order of the Padma awards?

(A) Padma Shri, Padma Bhushan, Padma Vibhushan
(B) Padma Shri, Padma Vibhushan, Padma Bhushan
(C) Padma Bhushan, Padma Shri, Padma Vibhushan
(D) Padma Vibhushan, Padma Bhushan, Padma Shri

Q.21
Which of the following acts introduced the system of dyarchy in the provincial administration in India?

(A) The Government of India Act of 1909
(B) The Government of India Act of 1947
(C) The Government of India Act of 1935
(D) The Government of India Act of 1919

Q.22
Which of the following statements is/are correct about the famous place Mawsynram? A. It is the wettest place on the earth. B. It is located in Meghalaya state of India.

(A) Only A is correct.
(B) Only B is correct.
(C) Neither A nor B is correct.
(D) A and B both are correct.

Q.23
Who was the founder of the Mauryan dynasty?

(A) Bindusara
(B) Ashoka
(C) Vikramaditya
(D) Chandragupta Maurya

Q.24
In which part of India is cool mountain climate observed?

(A) North
(B) South
(C) East
(D) West

Q.25
The Pharmacy (Amendment) Bill, 2023 was introduced in Lok Sabha on 3 August 2023, which amends the _____.

(A) Pharmacy Act, 1947
(B) Pharmacy Act, 1948
(C) Pharmacy Act, 1950
(D) Pharmacy Act, 1949

English Language and Comprehension 25 questions

Q.1
Select the option that correctly rectifies the underlined spelling error. The “accideent” on the highway caused a long traffic jam.

(A) accidnt
(B) accident
(C) acicdent
(D) acident

Q.2
Select the most appropriate option to fill in the blank. Accepting how wonderful you truly are and incorporating this _____ into your attitude and personality is perhaps one of the most difficult things to do in life.

(A) insight
(B) folly
(C) craziness
(D) glitch

Q.3
Select the option that correctly rectifies the underlined spelling error. The rainbow only appears after a “hurican”.

(A) hurrecain
(B) hurricane
(C) hurrican
(D) hurrecane

Q.4
Select the most appropriate meaning of the given idiom. A piece of cake

(A) Easy to do or achieve
(B) A cake that broke into pieces
(C) A cake which is sold as a piece
(D) A slice of cake for eating

Q.5
Select the most appropriate option to fill in the blank. Cultural heritage is the lifeblood of any _____ and serves as a link between generations.

(A) lethargy
(B) birthright
(C) believe
(D) civilisation

Q.6
Select the INCORRECTLY spelt word.

(A) Disgust
(B) Surrender
(C) Consicstent
(D) Trifle

Q.7
Select the most appropriate synonym of the word 'pursue' from the given sentence. Dr. Kalam's success in the field of science has inspired young Indians to dream big and chase those dreams with immense hard work.

(A) success
(B) immense
(C) inspired
(D) chase

Q.8
The following sentence has been split into four segments. Identify the segment that contains a phrasal verb error. Mrs. Premlatha asked / Malini to hand down / all the / payment slips.

(A) all the
(B) payment slips
(C) Mrs. Premlatha asked
(D) Malini to hand down

Q.9
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. My secretary is disposed of / discuss everything in great detail / before she begins / the actual work.

(A) the actual work.
(B) before she begins
(C) discuss everything in great detail
(D) My secretary is disposed of

Q.10

Select the most appropriate option to fill in the blank. Please listen to all announcements _____.

(A) generally
(B) easily
(C) hungrily
(D) attentively

Q.11

Select the most appropriate option to fill in the blank. He is _____ tired to finish his homework.

(A) to
(B) more
(C) too
(D) most

Q.12

Select the most appropriate synonym of the given word. Energy

(A) Speed
(B) Power
(C) Frailty
(D) strife

Q.13

The following sentence has been divided into four segments. One of them contains an error. Select the segment that contains the error from the given options. Neither Smith / nor Rose / were in / the classroom.

(A) the classroom
(B) Neither Smith
(C) nor Rose
(D) were in

Q.14

Select the sentence that has a grammatical error.

(A) He works as a software developer.
(B) She always eats healthy meals.
(C) The sun rises in the east.
(D) We watches movies every Friday night.

Q.15

Select the most appropriate synonym of the underlined word. The detective's job was to “discern” the truth from the web of lies.

(A) Perceive
(B) Ignore
(C) Confuse
(D) Fabricate

Q.16

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The workers had been repairing / the mall road in / town for / several days now.

(A) several days now
(B) town for
(C) The workers had been repairing
(D) the mall road in

Q.17

Select the most appropriate synonym of the given word. Ecstatic

(A) Happy
(B) Shocked
(C) Interested
(D) Angry

Q.18

Select the most appropriate synonym of the underlined word The project’s tight deadline and complex requirements made it a “gruelling” task for the team.

(A) Empowering
(B) Accessible
(C) Challenging
(D) Inclusive

Q.19

Select the most appropriate verb form to fill in the blank. Arun _____ his bicycle to school every day.

(A) rides
(B) rode
(C) riding
(D) ride

Q.20

What does it imply when people say to ‘read between the lines’?

(A) To avoid detailed information
(B) To skip initial and final lines in a paragraph
(C) To read with an ability to infer the meaning
(D) To read something fast

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to ____ 1 ___. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come ____ 2____ walks with us. This, however, we had to stop for, _____ 3____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4_____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____5_____.

Select the most appropriate option to fill in blank number 1.

(A) file
(B) flea
(C) read
(D) fly

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to ____ 1 ___. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come ____ 2____ walks with us. This, however, we had to stop for, _____ 3____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4_____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____5_____.

Select the most appropriate option to fill in blank number 2.

(A) in
(B) on
(C) at
(D) by

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to ____ 1 ___. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come ____ 2____ walks with us. This, however, we had to stop for, _____ 3____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4_____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____5_____.

Select the most appropriate option to fill in blank number 3.

(A) either
(B) neither
(C) never
(D) ever

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to ____ 1 ___. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come ____ 2____ walks with us. This, however, we had to stop for, _____ 3____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4_____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____5_____.

Select the most appropriate option to fill in blank number 4.

(A) peace
(B) pace
(C) piece
(D) space

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. My next pet, Ari, was a pigeon. He became convinced that he was not a bird at all and refused to ____ 1 ___. If he wanted to get on a table or a chair, he stood below it, cooing until someone picked him up. He would even try to come ____ 2____ walks with us. This, however, we had to stop for, _____ 3____ you carried him on your shoulder or else you let him walk behind. If you let him walk, you had to slow down your own _____ 4_____ to suit his, for should you get too far ahead, you would find him running after you, cooing frantically, his chest pouted out with _____5_____.

Select the most appropriate option to fill in blank number 5.

(A) indignation
(B) inclination
(C) intimation
(D) ignition

Answer key

Numerical and Mathematical Ability

Q.1 **(D)**	Q.2 **(C)**	Q.3 **(B)**	Q.6 **(B)**	Q.7 **(C)**	Q.8 **(B)**
Q.9 **(A)**	Q.11 **(B)**	Q.13 **(D)**	Q.14 **(B)**	Q.15 **(D)**	Q.16 **(B)**
Q.17 **(D)**	Q.18 **(D)**	Q.19 **(A)**			

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(C)**	Q.5 **(C)**	Q.6 **(A)**
Q.8 **(C)**	Q.9 **(A)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(C)**	Q.14 **(B)**
Q.15 **(B)**	Q.16 **(A)**	Q.17 **(B)**	Q.19 **(D)**		

General Awareness

Q.1 **(B)**	Q.2 **(D)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(B)**
Q.7 **(B)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(A)**	Q.11 **(D)**	Q.12 **(C)**
Q.13 **(C)**	Q.14 **(D)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(A)**	Q.20 **(D)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(A)**
Q.25 **(B)**					

English Language and Comprehension

Q.1 **(B)**	Q.2 **(A)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(D)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(B)**
Q.13 **(D)**	Q.14 **(D)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(C)**
Q.19 **(A)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(B)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(A)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (D) 4% loss

The first sack cost 19,200/1.2 = ₹16,000 and the second 19,200/0.8 = ₹24,000. Cost ₹40,000, sale ₹38,400: a loss of ₹1,600, or 4%.

Q.2 · Answer (C) ₹1,985.28

Successive discounts of 12% and 6%: ₹2,400 × 0.88 × 0.94 = ₹1,985.28.

Q.3 · Answer (B) 123

Take 2 kg, 1 kg and 5 kg: the 8 kg of mixture is worth 8 × 96 = ₹768, so 86 + 67 + 5x = 768, 5x = 615 and x = ₹123 a kg.

Q.6 · Answer (B) 65.40

The total rises by 88 − 68 = 20, so the average rises by 20/50 = 0.4: 65 + 0.4 = 65.40.

Q.7 · Answer (C) 24

In 12 seconds the thief gets 18 × 12 = 216 m ahead. The policeman's 864 m takes as long as the thief's 864 − 216 = 648 m, i.e. 648/18 = 36 s, so his speed is 864/36 = 24 m/s.

Q.8 · Answer (B) 87

The numbers are 6x, 7x and 3x, so their LCM is 42x = 3654 and x = 87. As 6, 7 and 3 have no common factor, the HCF is x = 87.

Q.9 · Answer (A) 10 sec

Moving towards each other, the relative speed is 62 + 46 = 108 km/h = 30 m/s. They must cover 148 + 152 = 300 m: 300/30 = 10 seconds.

Q.11 · Answer (B) 1470

SI = 3,500 × 7 × 6/100 = ₹1,470.

Q.13 · Answer (D) 15 : 4

(1/5) × boys = (3/4) × girls, so boys : girls = (3/4) : (1/5) = 15 : 4.

Q.14 · Answer (B) 38

Let the ages be 3x and x. (3x − 5)/(x − 5) = 5 gives 3x − 5 = 5x − 25, so x = 10 and A is 30. In eight years A will be 38.

Q.15 · Answer (D) 14,975

Expenditure = 3 × 8,600 + 4 × 9,750 + 5 × 9,920 = 25,800 + 39,000 + 49,600 = ₹1,14,400. Income = 1,14,400 + 65,300 = ₹1,79,700, or ₹14,975 a month.

Q.16 · Answer (B) ₹992

P × (1 + 5 × 6/100) = 806, so P = 806/1.3 = ₹620. At 12% for 5 years: 620 × (1 + 0.6) = ₹992.

Q.17 · Answer (D) 1050

SI = 3,000 × 7 × 5/100 = ₹1,050.

Q.18 · Answer (D) ₹7,160

A + B = 7,180, B + C = 8,500 and A + C = 13,000. Adding, 2(A + B + C) = 28,680, so A + B + C = 14,340 and C = 14,340 − 7,180 = ₹7,160.

Q.19 · Answer (A) 200

Going from a 20% loss to a 30% profit is a swing of 50% of the cost price, which is ₹100: so the cost price is ₹200.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) UWY
First letters move 5 places forward (A, F, K, P, so next U), and within each cluster the letters go up by two (ACE, FHJ): UWY.

Q.2 · Answer (D) Third to the left
Going round to the right from E the seats are E, C, A, B, F, D, G (C on E's immediate right, D third to C's left, A between C and B, and F, who is not next to E, beside D). Counting to A's left: C, E, G, so G is third to the left.

Q.3 · Answer (B) Two
Going round to the right from E the seats are E, C, A, G, D, F, B (A second to E's right, D two seats beyond A, B third to G's right with F beside it). Counting to E's left: B, F, then D, so two people sit between them.

Q.4 · Answer (C) Neither conclusion I nor II follows
The chickens that are hens are not eggs, but nothing is said about any other chickens, so I does not follow. Nothing says every hen is a chicken, so II does not follow either.

Q.5 · Answer (C) TRXU
Each letter moves 7 places forward: H → O, F → M, L → S, I → P, and KIOL → RPVS likewise. So MKQN → TRXU.

Q.6 · Answer (A) -192
Each term is the one before multiplied by –2: +96 × (–2) = –192, and –192 × (–2) = +384.

Q.8 · Answer (C) 64
The code is 8 times the number of letters: COLONY (6 letters) → 48 and INTERCHANGE (11) → 88. MOUNTAIN has 8 letters: 64.

Q.9 · Answer (A) Both conclusions I and II follow.
A universal negative works both ways: 'No bus is a car' means no car is a bus (I), and 'No car is a truck' means no truck is a car (II). Both follow.

Q.10 · Answer (D) 136
After interchanging the signs: 39 × 3 + 24 – 60 ÷ 12 = 117 + 24 – 5 = 136.

Q.11 · Answer (C) 6
N is the only letter in both words and 6 the only digit in both codes, so N = 6.

Q.12 · Answer (C) NQ26
Both letters move 5 places back and the number rises by 3: JM12 → EH15 and QT34 → LO37. So SV 23 → NQ26.

Q.14 · Answer (B) 180
The second number is 10 times (the first minus 4): 10 × 3 = 30 and 10 × 10 = 100. So 22 → 10 × 18 = 180.

Q.15 · Answer (B) lk
'I' is the only word in both sentences and 'lk' the only code in both, so I = lk.

Q.16 · Answer (A) UKPG
The letters move –2, +2, –2, +2: CAKE → ACIG and GOLD → EQJF. So WIRE → UKPG.

Q.17 · Answer (B) (13, 53, 20)
In each set the middle number is the sum of the other two plus 20: 4 + 11 + 20 = 35 and 1 + 61 + 20 = 82. Likewise 13 + 20 + 20 = 53.

Q.19 · Answer (D) 3
After interchanging the signs: 5 × 6 ÷ 3 + 4 – 11 = 10 + 4 – 11 = 3.

General Awareness

Q.1 · Answer (B) Ladakh
Losar, the Buddhist new-year festival, is celebrated in Ladakh.

Q.2 · Answer (D) Smriti Mandhana
Smriti Mandhana captained Royal Challengers Bangalore in the first Women's Premier League in 2023.

Q.3 · Answer (A) Samajwadi Party
Shafiqur Rahman Barq, MP for Sambhal in Uttar Pradesh, belonged to the Samajwadi Party.

Q.4 · Answer (A) Decorating the walls with floral designs made of semi-precious stones
Pietra dura is the inlay of cut and polished semi-precious stones in marble to form floral designs, as on the Taj Mahal.

Q.5 · Answer (C) $NaHCO_3$
Sodium bicarbonate (baking soda) is $NaHCO_3$. Na_2CO_3 is sodium carbonate (washing soda), NaOH sodium hydroxide and NaCl common salt.

Q.6 · Answer (B) Seven
India shares land borders with seven countries: Pakistan, Afghanistan, China, Nepal, Bhutan, Bangladesh and Myanmar.

Q.7 · Answer (B) Poverty and Un-British Rule in India
Dadabhai Naoroji's 'Poverty and Un-British Rule in India' (1901) set out his drain theory: that British rule was draining India's wealth.

Q.8 · Answer (A) curcumin
Curcumin gives turmeric its yellow colour; capsaicin is the hot compound in chillies.

Q.9 · Answer (A) Sangama dynasty
Harihara and Bukka of the Sangama dynasty founded the Vijayanagara kingdom in 1336; the Saluva, Tuluva and Aravidu dynasties followed.

Q.10 · Answer (A) Roller skating
Sanjana Bathula is an Indian roller speed skater; she was part of India's women's 3000 m relay squad at the Hangzhou Asian Games in 2023.

Q.11 · Answer (D) Gujarati
K. M. Munshi wrote 'Patan ni Prabhuta' in Gujarati; 'The Glory of Patan' is its English translation.

Q.12 · Answer (C) March-April
The rabi season begins with the onset of winter in October-November and ends in March-April; its cool weather suits wheat, gram and mustard.

Q.13 · Answer (C) 1 January 1970
SSC's key gives 1 January 1970, the date usually given for the abolition of Punjab's Legislative Council, its upper house, under the Punjab Legislative Council (Abolition) Act, 1969.

Q.14 · Answer (D) Dadra and Nagar Haveli
In the 2011 Census Dadra and Nagar Haveli had the lowest literacy rate of the Union Territories, about 76%.

Q.15 · Answer (A) Mohiniyattam
Kanak Rele, who founded the Nalanda Dance Research Centre in Mumbai, was a leading exponent of Mohiniyattam.

Q.16 · Answer (C) Ustad Amir Khan
The Indore gharana of Hindustani vocal music was founded by Ustad Amir Khan.

Q.17 · Answer (D) lycopene
Lycopene, a red carotenoid pigment, colours ripe tomatoes.

Q.18 · Answer (A) They are durable.
Capital goods, such as machinery and buildings, are durable: they are used in production over many years.

Q.19 · Answer (A) IV A
The 42nd Amendment (1976) added Part IV A, whose Article 51A lists the Fundamental Duties.

Q.20 · Answer (D) Padma Vibhushan, Padma Bhushan, Padma Shri
Padma Vibhushan ranks highest, then Padma Bhushan, then Padma Shri.

Q.21 · Answer (D) The Government of India Act of 1919
The Government of India Act, 1919 (the Montagu-Chelmsford Reforms) introduced dyarchy in the provinces; the 1935 Act ended it there.

Q.22 · Answer (D) A and B both are correct.
Mawsynram, in Meghalaya's East Khasi Hills, receives the highest average annual rainfall of any place on earth, so both statements are correct.

Q.23 · Answer (D) Chandragupta Maurya
Chandragupta Maurya founded the Mauryan dynasty in the late 4th century BC; Bindusara and Ashoka were his son and grandson.

Q.24 · Answer (A) North
The Himalayan regions of the north have a cool mountain climate, as temperatures fall with altitude.

Q.25 · Answer (B) Pharmacy Act, 1948
The Bill amends the Pharmacy Act, 1948, which regulates the profession of pharmacy, so that pharmacists registered under the Jammu and Kashmir Pharmacy Act, 2011 can be registered under it.

English Language and Comprehension

Q.1 · Answer (B) accident
The correct spelling is 'accident'.

Q.2 · Answer (A) insight
Accepting how wonderful you are is an 'insight', an understanding, to build into your attitude; 'folly', 'craziness' and 'glitch' do not fit.

Q.3 · Answer (B) hurricane
The correct spelling is 'hurricane'.

Q.4 · Answer (A) Easy to do or achieve
'A piece of cake' is something very easy to do.

Q.5 · Answer (D) civilisation
Cultural heritage links the generations of a 'civilisation'; 'believe' is a verb, and 'lethargy' and 'birthright' do not fit.

Q.6 · Answer (C) Consicstent
The correct spelling is 'consistent'.

Q.7 · Answer (D) chase
To 'pursue' a dream is to 'chase' it.

Q.8 · Answer (D) Malini to hand down
To 'hand down' is to pass something on to a later generation; slips are 'handed in' (submitted) or 'handed over'.

Q.9 · Answer (D) My secretary is disposed of
'Disposed to' (inclined to) takes a verb: 'My secretary is disposed to discuss everything'. 'Disposed of' means got rid of.

Q.10 · Answer (D) attentively
Announcements should be listened to 'attentively', with full attention.

Q.11 · Answer (C) too
'Too … to' says a quality is excessive for a result: 'too tired to finish'.

Q.12 · Answer (B) Power
'Energy' is the 'power' to act or work.

Q.13 · Answer (D) were in
With 'neither … nor' the verb agrees with the nearer subject, here the singular 'Rose': 'Neither Smith nor Rose was in the classroom.'

Q.14 · Answer (D) We watches movies every Friday night.
'We' takes the plural verb: 'We watch movies every Friday night.'

Q.15 · Answer (A) Perceive
To 'discern' is to 'perceive' or recognise clearly.

Q.16 · Answer (C) The workers had been repairing
An action that began in the past and is still going on 'now' takes the present perfect continuous: 'The workers have been repairing'.

Q.17 · Answer (A) Happy
'Ecstatic' means extremely 'happy'.

Q.18 · Answer (C) Challenging
A 'gruelling' task is exhausting and demanding: 'challenging'.

Q.19 · Answer (A) rides
A habitual action takes the simple present, and the singular subject 'Arun' takes 'rides'.

Q.20 · Answer (C) To read with an ability to infer the meaning
To 'read between the lines' is to infer a meaning that is not stated openly.

Q.21 · Answer (D) fly
Convinced that he was not a bird, Ari refused to 'fly': he waited to be lifted onto tables and chairs instead.

Q.22 · Answer (B) on
One goes 'on' a walk: he would even try to come on walks with us.

Q.23 · Answer (A) either
'Either' pairs with 'or else' to offer two choices: either you carried him or else you let him walk behind.

Q.24 · Answer (B) pace
You slowed your own 'pace', your walking speed, to suit his; 'peace', 'piece' and 'space' make no sense here.

Q.25 · Answer (A) indignation
Left behind, he ran after you with his chest puffed out in 'indignation', anger at being treated unfairly.

This paper online, with tap-to-reveal answers: https://parikshasaathi.com/exams/ssc-mts/question-papers/mts-2024-30-sep-2024-shift-1/
Every solved SSC MTS paper: https://parikshasaathi.com/exams/ssc-mts/question-papers/
Daily current affairs PDF, free, every morning on Telegram: t.me/parikshasaathiapp
Questions and official answers: the Staff Selection Commission (SSC). Explanations: drafted with AI assistance by Pariksha Saathi and checked against the official answers. Pariksha Saathi is not affiliated with any exam body. Spotted an error? Write to sriwastava2@gmail.com.